

# SERGEY CHERNYAKOV



## AI Engineering Lead · Principal Backend Engineer · AI Automation Architect

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Age: 47 · Male

GitHub: github.com/sergeychernyakov

Location: Wrocław, Poland · Remote / EU (open to relocation, hybrid 1–2 days/week OK)

## PROFESSIONAL SUMMARY

Senior software engineer, backend architect and AI engineering lead with **20+ years** of commercial experience in software development, automation and scalable backend systems.

Started professional career in 2004 and have worked across media platforms, SaaS products, fintech, CRM systems, blockchain infrastructure, AI automation and enterprise integrations.

Strong expertise in **Python (FastAPI / Django / Flask), Ruby on Rails 4.x → 7.x, Node.js / TypeScript, PHP / Laravel, PostgreSQL, Docker** and distributed backend systems. Designs production APIs, automation pipelines, AI-driven development workflows and multi-agent AI systems.

Current focus: **AI engineering, AI automation and LLM integrations** for production environments — OpenAI APIs, AI agents, Model Context Protocol (MCP) servers, AI-assisted SDLC. Top Rated on Upwork — **18,000+ hours, 71+ long-term contracts, \$700K+ earnings, 98% Job Success**.

Available as **B2B contractor via Deel.com**.

## KEY SKILLS

- **Languages:** Python · Ruby · PHP · JavaScript / TypeScript · SQL · Solidity · (learning Go, Rust)
- **Backend frameworks:** FastAPI · Django · Flask · Ruby on Rails · Laravel · Node.js / Express
- **AI / ML:** OpenAI API · Anthropic Claude API · LLM integration · Multi-agent AI systems · RAG · Vector DBs · OCR / Vision LLMs · MCP (Model Context Protocol) servers · AI-assisted SDLC
- **Data & storage:** PostgreSQL · MySQL · ClickHouse · Redis · Kafka · MongoDB · ETL & data pipelines
- **Infra & DevOps:** Docker · Linux · GitHub Actions · GitLab CI · AWS (EC2 / S3 / RDS) · Nginx · Prometheus · Grafana
- **Architecture:** REST · GraphQL · Microservices · WebSockets · Event-driven systems · OpenTelemetry distributed tracing · Packwerk-enforced module boundaries
- **Integrations:** Stripe · PayPal · Braintree · Recurly · Bluesnap · YooKassa · Salesforce · Mailchimp · Drip · Twilio · Telegram & WhatsApp Bot APIs · Selenium-based RPA
- **Blockchain:** Ethereum · Solana · Web3.py · ethers.js · Solidity smart contracts · Flashbots · Uniswap V2/V3 · DEX arbitrage · MEV

## PROFESSIONAL EXPERIENCE

### AI Engineering Lead / AI Automation Architect — Sabermine (United Kingdom, Remote)

November 2024 – November 2025

Led AI engineering and automation direction for internal and client systems.

- Designed and operated Python-based **ETL and automation pipelines** for data ingestion, transformation and downstream analytics
- Built **multi-agent AI systems** for development, QA, documentation and DevOps automation
- Backend services on **Python, FastAPI, PostgreSQL**; Docker-based deployment, CI/CD via GitHub Actions
- Integrated **OpenAI APIs / LLM workflows** into production for OCR, document classification and content generation
- Built internal tooling to improve data quality, observability and reproducibility of AI pipelines
- Improved development speed and operational efficiency **5–10x** via AI-assisted SDLC

### Senior / Principal Backend Engineer — Upwork (International Clients, Remote)

June 2012 – June 2025 (Top Rated)

**18,000+ hours · 71+ contracts · \$700K+ earnings · 98% Job Success · 5★ feedback**. Worked with international clients from the US, UK, EU and Australia. Long-term enterprise projects across SaaS, fintech, marketing automation, e-commerce, AI and Web3.

**Recent (2024–2026) selected work:**

- **AUROSPEACE** (`aurospace`) — production-style Rails 7 + React payment workflow showcase. Modular monolith with AASM state-machine orders, ledger-based money movement, Yookassa payment processing with webhooks, Packwerk-enforced module boundaries, dry-monads service layer, OpenTelemetry distributed tracing, ActiveAdmin, RSpec.
- **Solana Pulse Bot** (`solana_pulse_bot`) — real-time ML scoring pipeline for Solana memecoin launchpads. 8 XGBoost heads (entry / entry @T+30 / regression / entry-timing / survival / SL+TP+max\_hold quantile heads), **82 features per token** (schema v21), **100% backtest-live decision parity** verified on 300+ tokens, config-hash drift guards, 5-seed feature-stability protocol.
- **МаркетПульт** (private) — multi-tenant Rails SaaS for managing remote outbound sales staff: daily plans, activity proofs, payroll calculation, role-based access.
- **Спец Ресурс** (private) — Laravel 12 + React 18 + Inertia.js corporate portal for outsourced personnel management; 5-role access model (Main Admin / Manager / Store / Employee + Senior Manager); mobile UI for field staff; PostgreSQL 16; Prometheus + Grafana / Zabbix monitoring.
- **ImWriter Wizard** (private) — Rails 7.2 + Hotwire (Turbo + Stimulus) AI book constructor; PostgreSQL with UUID PK + JSONB; Devise + Pundit; OpenAI-powered chapter generation, character creation, reviews; i18n (en/ru/es/zh).
- **Umetex** (`umetex`) — Django + ChatGPT medical document translator preserving layout (tables, images), Cyrillic font handling, admin panel with predefined-phrase consistency.
- **Multi-agent MCP servers** (`aq_ai_agent`, `weather_mcp_agent`) — FastAPI + Model Context Protocol via Server-Sent Events; legacy SOAP CRM (Aquarium) exposed as agent-callable MCP tools.
- **DeFi infrastructure**: DEX arbitrage with Balancer flashloans + Multicall3 (95x perf, 4,750 combos/min, Uniswap V2/V3, Sushiswap, Pancakeswap, Doorswap; private TX via Flashbots; MongoDB; cron orchestration); Solana wallet bundlers (TypeScript + Raydium/Serum); Ethereum token launch+bundle (Node.js + Flashbots); private-TX sender (Web3.py); Uniswap V2 token-sell bot with gas estimation.
- **Voice & chat AI**: OpenAI-powered Asterisk voice assistant (STT + TTS + email summaries); WhatsApp + GPT bot with MongoDB session persistence (`instreamatic`); multi-modal Telegram bots (`aihelper` — Translator + Diet with payments).
- **RPA / Selenium automation**: UK Financial Ombudsman Service form submission (`fos_submission_automator`, Selenium + Pydantic + anti-detection); Barclays SecureMail end-to-end pipeline (`securemail_automator`); FastDox API v2 bulk PDF extraction — 32,000+ documents into Azure Blob Storage (`fastdox_exorcist`); Blackhorse DCA Checker form bot.
- **Computer Vision**: Endangered-species video analytics with Yandex Maps + RBAC (`redeye`); surveillance camera management for shopping malls / industrial sites (`surveye`); neural-network meter reading vs. manual readings (`frisson`); GPT-Vision photo transcription web app (`tagman`).

#### Earlier long-term engagements (2012–2024):

- **Clickfunnels** (USA, 2012–2017) — Backend engineering on high-volume marketing-automation SaaS. Built data flows, payment / event pipelines and integrations with **Salesforce, Mailchimp, Drip, Aweber, Constant Contact, GoToWebinar, Google Ads / Adwords API, Stripe, Braintree, Recurly, Twilio**. Stack: Ruby on Rails, MySQL, Kafka, Redis, Docker. Reference: *"Seriously one of the best hires we've ever made on Upwork. Sergey has continued to exceed our expectations."* — Ryan Montgomery, COO.
- **Gleam.io** (Australia, 2015–2016) — Backend integrations with social and audience APIs (Facebook, Twitter, Instagram, Pinterest, YouTube, Tumblr, SoundCloud) for SaaS competitions platform. Stack: Ruby on Rails + Angular, REST APIs, AWS hosting.
- **Lawdingo** (USA, 2010–2012) — Lawyer marketplace with phone-based service and automated payments. Stack: Ruby on Rails, MySQL, AWS (EC2 + Apache+Passenger), Sidekiq, Solr (search), Stripe, Twilio.
- **Claricclaim** (UK, insurance) — Laravel backend + Google OCR + Zoho integration; claim automation, document workflow, audit logging.
- **DA Languages** (UK), **Saferkid**, **Pashadelic** — various long-term engagements.

#### Senior Fullstack Engineer — Fit2you (Italy, insurance)

April 2022 – January 2024

- Internal enterprise systems for an Italian insurance company (dealer / vendor / client portals)
- Migrated all company applications from **Rails 4.1** → **Rails 7.1** (5 legacy apps) with SonarQube quality gates and 80% test coverage
- Supported and developed **all 8 of the company's applications** end-to-end — feature work, maintenance and incident fixes; one of them was a **Node.js + React** app, the rest on Ruby on Rails
- Stack: Ruby, Ruby on Rails, Node.js / Express, TypeScript, Bootstrap, **PostgreSQL**

#### Senior Ruby on Rails / PHP Engineer — Skykillers Digital Agency (Kyiv, Ukraine)

August 2011 – June 2012

- Lead developer at a digital advertising agency; team management + programming
- Campaign and brand sites for **Colgate, Danone, Nicorette, Visine, Davidoff, Johnson's Baby**, including Facebook campaign apps and promo platforms

### Senior PHP Developer — ITC Publishing (Ukrainian media holding)

*June 2009 – August 2011*

- High-traffic media properties (hotline.ua, itc.ua, dpk.com.ua, video.itc.ua); CMS on Zend Framework, MySQL
- Internal CRM and editorial workflow automation

### Earlier — Media Holding RBC-Ukraine, Obozrevatel, Quasar-Micro Techno (KM Core), PR-media

*2004 – 2009*

Top Ukrainian media portals (rbc.ua, utro.ua, loveplanet.ua, obozrevatel.com), eTutor virtual learning platform. PHP, MySQL, Zend Framework, web analytics, SEO.

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## SELECTED OPEN-SOURCE / PORTFOLIO PROJECTS

- **aurospace** (Rails 7 + React + PostgreSQL + Sidekiq) — Modular monolith fintech showcase. [github.com/sergeychernyakov/aurospace](https://github.com/sergeychernyakov/aurospace)
- **solana\_pulse\_bot** (Python + XGBoost + PostgreSQL) — Real-time ML scoring with 100% backtest-live parity. [github.com/sergeychernyakov/solana\\_pulse\\_bot](https://github.com/sergeychernyakov/solana_pulse_bot)
- **umetex** (Django + ChatGPT) — Medical document translation with layout preservation. [github.com/sergeychernyakov/umetex](https://github.com/sergeychernyakov/umetex)
- **aq\_ai\_agent** (FastAPI + MCP + SOAP CRM) — Legacy CRM exposed as agent-callable MCP tools via SSE. [github.com/sergeychernyakov/aq\\_ai\\_agent](https://github.com/sergeychernyakov/aq_ai_agent)
- **private\_transaction\_sender** (Web3.py + Flashbots) — MEV-protected Ethereum private TX submission. [github.com/sergeychernyakov/private\\_transaction\\_sender](https://github.com/sergeychernyakov/private_transaction_sender)
- **asterisk** (Python + OpenAI Whisper/TTS + Asterisk PBX) — Voice AI assistant on legacy telephony.

**Plus 20+ private repos** in similar stacks (multi-tenant SaaS, Laravel 12 + React + Inertia, AI book constructor with Hotwire, DEX arbitrage with flashloans, UK financial-services form automation, surveillance video analytics) available to discuss on call.

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## EDUCATION

**Bachelor's Degree** — Don State Technical University, 1999

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## LANGUAGES

Ukrainian — Native · English — C1 (Fluent professional) · Polish — Learning (A1)

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## ENGAGEMENT

- **B2B contractor** — invoicing via **Deel.com** (current setup; open to other platforms)
- **Format:** Remote (preferred) · Hybrid OK (1–2 days/week)
- **Location / mobility:** Wrocław, Poland · Open to relocation within EU · Visa sponsorship welcome for non-Poland EU
- **Notice period:** ~1 month
- Remote-first engineer experienced in distributed international teams and long-term enterprise projects.
- Strong practical engineering mindset with focus on production systems, scalability, maintainability and automation.
- Actively using AI-assisted development workflows, AI agents and automation tooling in real production environments.